HARVARD UNIVERSITY

Harmful Brain Activity Classification

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Abstract

This project aims to develop an automated system for the classification of seizures and other harmful brain activities using electroencephalography (EEG) data. The goal is to address the challenges associated with the manual process of EEG monitoring, which is prone to errors and inconsistencies among neurologists. By leveraging machine learning techniques, the project seeks to improve the accuracy and efficiency of seizure detection. This project is part of a finished Kaggle competition that can be found here.

Significant progress in automated EEG pattern classification has been made through the application of classic machine learning algorithms and, more recently, deep learning approaches. Our approach further enhances these methodologies by integrating image-based and time-series classifiers within a single ensemble model to leverage both spatial and temporal information from EEG data.

The dataset utilized includes 11,138 instances of annotated EEG recordings. This project evaluates the model's performance using the Kullback-Leibler Divergence method and aims to rank among the top performers in the competition by minimizing the difference between the classifications assigned by the model and that made by experts who annotated the recordings. The results indicate promising directions for future research and potential clinical applications in real-time EEG monitoring systems.

Related works

Automated classification of EEG patterns for seizure detection began to gain significant attention in the early 2010s, utilizing simpler algorithms such as decision trees and K-Nearest Neighbors (KNNs). Cherian et al. (2022) and Siddiqui et al. (2020) described early efforts where features were manually extracted using statistical methods like Fourier Transforms for frequency analysis, which were then fed into these models.

In recent developments, deep learning methods have been increasingly applied to this field. Hu et al. (2020) introduced a deep bidirectional Long Short-Term Memory (Bi-LSTM) network for classifying EEG patterns, a technique suited for time-series data due to its ability to capture temporal dependencies. Their architecture employs two LSTM networks with opposite data flow directions, enabling the model to utilize information both before and after a given point in time for improved classification accuracy. The model, which processed statistical features like standard deviation, mean, and others, achieved high sensitivity (93.6%) and specificity (91.9%) on the CHB-MIT dataset, a benchmark in EEG seizure detection research.

Furthermore, attempts at classification using CNNs have been tried to reduce the burden of manually performing feature engineering on the EEG data. Usman et al. (2020) developed a CNN to automatically extract features from raw EEG data. The extracted features were then fed into an SVM (Support Vector

Machine) to classify the EEG data and detect seizures. This method achieved 92.7% sensitivity and 90.8% specificity in the CHB-MIT dataset.

Jing et al. (2023) presented one of the latest approaches to classify EEG patterns for seizure detection. The authors on this paper are responsible for the Kaggle competition previously mentioned, thus this work is extremely relevant to this project. In their work, they developed SPaRCNet, a DenseNet based CNN. Their approach is to transform EEG data into spectrogram images, capitalizing on CNNs' ability to process visual information. This way, they have turned the issue of harmful activity detection into an image classification problem, and they can leverage state-of-the-art models such as DenseNet.

The objective of this project is to improve on the work started by Jing et al. (2023). Converting EEG to spectrograms allows one to use image classification techniques. However, dynamic temporal dependencies in are lost when converting the EEG data to the frequency domain. We propose developing an ensemble of models, leveraging both the strengths of CNNs to classify spectrograms, and time-series based models such as WaveNet to classify the raw EEG data. Similar approaches have been developed successfully in other areas. As an example, Vankdothu et al. (2022) developed a combined CNN-LSTM (Long Short-Term Memory) approach to classify brain tumor images, where the hybrid model reached 92% accuracy, while each individual model alone had lower performance (CNN: 89%, LSTM: 90%).

Method

Model

The core of our proposed method involves the development of an ensemble model that integrates image-based and time-series classifiers. This design is inspired by the innovative CNN-LSTM hybrid model discussed by Vankdothu et al. (2022). By combining these two types of classifiers, we aim to enhance the model's ability to interpret and classify complex EEG data, improving upon the foundational methods used by Jing et al. (2023).

Dataset

The dataset utilized in this project is provided on the Kaggle platform and consists of 11,138 EEG recordings, each containing 10 minutes of data. These recordings are accompanied by corresponding spectrograms and have been annotated by 3 to 20 experts. The experts' annotations serve to classify each EEG into one of six recognized patterns: seizure (SZ), generalized periodic discharges (GPD), lateralized periodic discharges (LPD), lateralized rhythmic delta activity (LRDA), generalized rhythmic delta activity (GRDA), or "other". For illustrative purposes, Figures 1a and 1b showcase examples of these data annotations. The dataset is available for download here.

Figure 1a – An example where all twenty experts voted to classify the pattern as a Seizure (SZ).

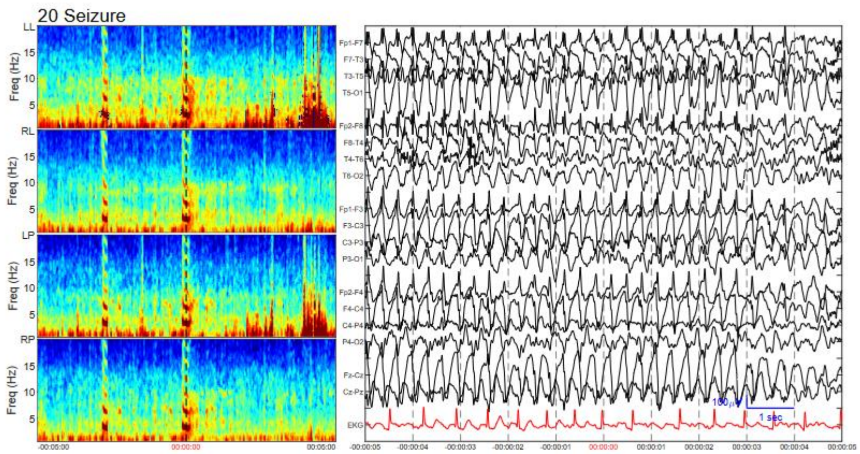
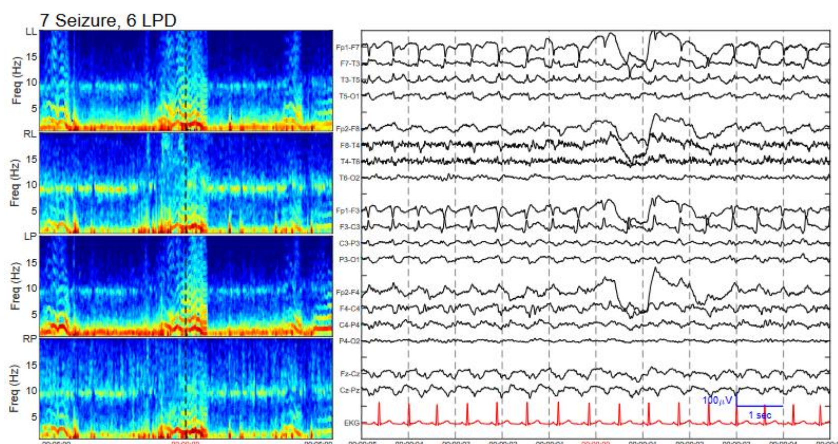


Figure 1b – An example where seven experts voted to classify the pattern as a Seizure (SZ) and six voted to classify it as Lateral Periodic Discharge (LPD).



<u>*Measuring results*</u>

To evaluate our model's performance, we convert the expert votes into discrete probability distributions. For instance, if seven out of thirteen experts classify a recording as a seizure and the remaining six as a lateralized periodic discharge (as shown in Figure 1b), the distribution would reflect a 54% probability for seizure and 46% for LPD. The model's effectiveness is then quantified using the Kullback-Leibler Divergence (KL), which measures the distance between the probability distributions predicted by our model and those determined by expert consensus. The Kullback-Leibler equation to measure the distance between two discrete probability distributions P and Q is shown below.

$$D_{\mathrm{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \, \log\left(\frac{P(x)}{Q(x)}\right)$$

Code base

Our project leverages a starter notebook released by Chris Deotte, a Kaggle user. This notebook implements a convolutional neural network using EfficientNetB0 to classify spectrograms. It has been adapted from TensorFlow to PyTorch and used as the basis of our initial model development. The code base can be found here.

Definition of success

It is unclear if the dataset used in the competition is the same as used by Jing et al. (2023), so directly comparing our results is not feasible. Therefore, we define our project's success based on improvements over the baseline provided by the code base, which has a KL score of 0.43. Success will be considered as achieving a KL score between 0.25 and 0.34, placing us within the top 20% of competitors on the Kaggle leaderboard as of March 8th.

Implementation

The steps taken to develop the project are seen below.

Phase 1 – Development and Evaluation of Separate Models: The initial phase of our project focused on the development and independent evaluation of two distinct model types: image-based and time-series-based models. This stage was crucial for establishing a solid understanding of each model's unique strengths and limitations. Different architectures, such as EfficientNet and ResNet, were explored in this stage.

Phase 2 – Development of the Hybrid Model: The second phase involved the development of the ensemble model combining strengths of both models to leverage both spatial and temporal information present in the EEG data. To integrate the outputs of the two models, we opted for a weighted average of the outputs due to the simplicity and effectiveness of this method.

Phase 3 – Fine-Tuning: In the final phase, our efforts were concentrated on optimizing the performance of the ensemble model through systematic hyperparameter fine-tuning. We employed a random search strategy to navigate the hyperparameter space, which allowed us to identify the most effective settings for maximizing the model's accuracy.

The links to the final version of the code are found below.

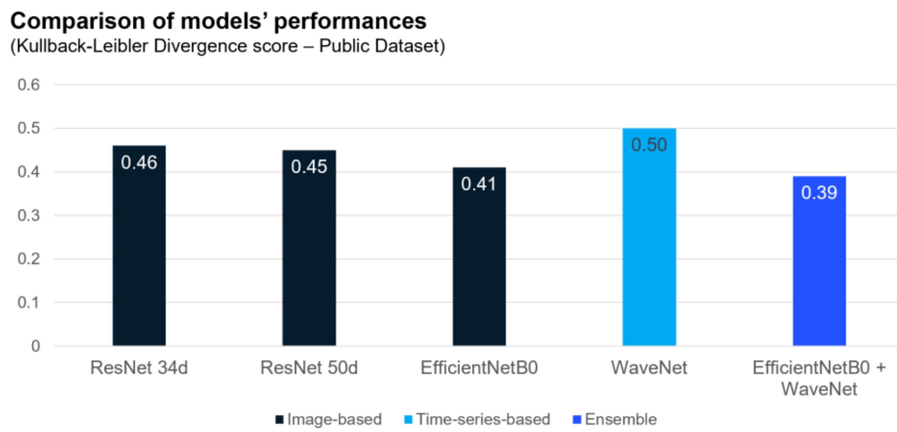
- **Code base:** available here.
- **Finalized ensemble model:** available here.

Results

In the initial phase of our project, various individual models were evaluated, with a particular focus on their Kullback-Leibler (KL) divergence scores. EfficientNetB0 emerged as the standout among all models.

Due to computational constraints within the Kaggle platform, more complex iterations of EfficientNet, featuring additional layers, could not be explored. The ensemble model, which combined a weight of 0.75 for EfficientNetB0 and 0.25 for WaveNet, outperformed all individual models (see Figure 2 for the scores of the different models used).

Figure 2 – Scores achieved by the different models used.



Analyzing results in this Kaggle competition presented unique challenges. Firstly, the absence of disclosed labels for the test dataset complicates analyses, limiting our ability to precisely identify the model's weaknesses. Secondly, the probabilistic nature of the data labels renders typical performance assessments like confusion matrices less relevant than when dealing with discrete categories.

To navigate these obstacles, we employed cross-validation on the training data, partitioning it into five folds to ensure robustness and generalizability of our findings. The distribution of labels within the training data is depicted in Figure 3. Furthermore, to further analyze the comparative strengths of the selected models for the ensemble, we conducted a detailed error analysis. Traditional confusion matrices were replaced by conditional probability matrices, which better suit the probabilistic label structure. The matrices are shown in Figure 4.

Figure 3 – Distribution of classes in the training dataset.

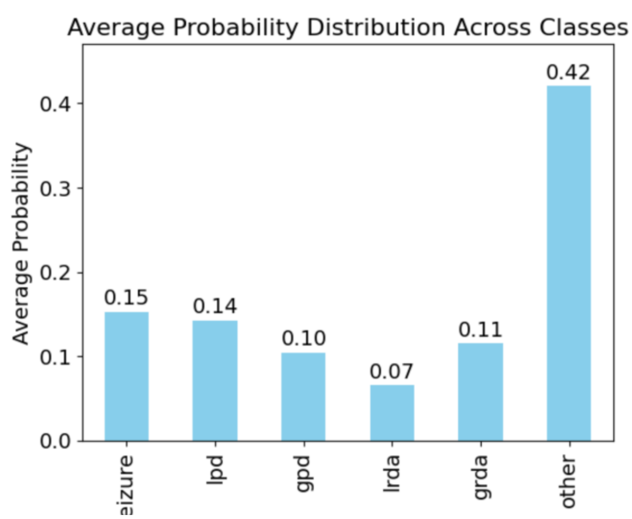
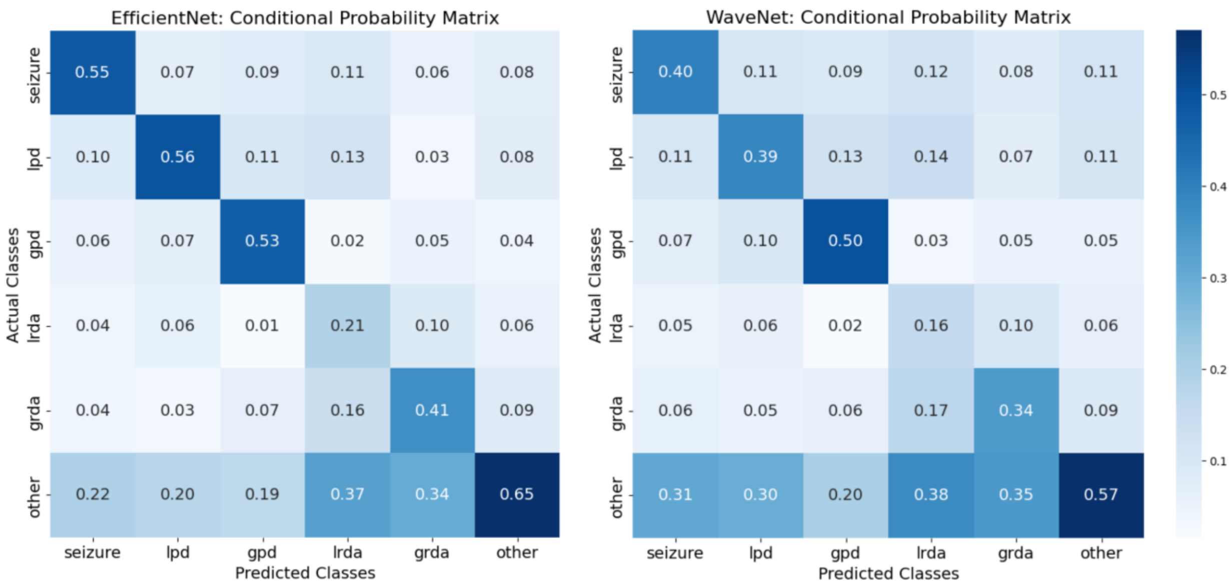


Figure 4 – Conditional probability matrices of the two models used in the ensemble.



Discussion

As depicted in Figure 4, these matrices revealed an intriguing aspect of model performance: although EfficientNet consistently outperformed WaveNet across all classes, the ensemble model, which combines these two, demonstrated superior overall effectiveness. The superior performance of the ensemble hints at underlying complexities within the data that are not captured by straightforward class-level analysis. It suggests that certain data patterns, which may not lead to high accuracy for any single class when models are considered individually, are better captured by the integration of different modeling approaches. This nuanced handling of data intricacies by the ensemble model underscores the potential benefits of model diversification, particularly in scenarios where data labels are probabilistic and potentially overlapping in feature space.

A notable difficulty encountered by both models was accurately classifying recordings labeled as “Other.” This category was particularly problematic due to its significant representation in the dataset and the inherent ambiguity in its definition—comprising patterns that experts could not categorize definitively. This suggests that the “Other” category likely encompasses a diverse array of patterns, rather than a homogeneous group. Interestingly, this might indicate that the models are effectively recognizing distinct patterns within this category that were not explicitly identified by the experts. As we regard expert annotations as the gold standard, this observation raises important considerations about the potential of machine learning models to discern subtle patterns beyond human expert classification.

The KL score achieved by the ensemble (0.39) places it at the median of competition participants, specifically 1,369th out of 2,767 competitors. While this represents a significant advancement over the initial code base (0.39 vs. 0.43), which is an improvement of approximately 10%, it still falls short of the project's initial objective to rank in the top 20% of competitors. Throughout the competition, other competitors managed to rank higher through leveraging more effective code bases released by other participants, adopting diverse strategies which we will discuss further below.

The competition concluded on April 9th, with the highest score recorded at 0.22. The methodologies employed by the top-performing models in the competition are detailed in Table 1. This analysis aims to highlight potential strategies that could have further enhanced the performance of our project.

Table 1 – Approaches used by the winners of the competition.

Rank	User/Team	High-level description
1	Team Sony	**Model:** Ensemble of four different models, all using image-based classifiers. **Approach:** The focus was on data pre-processing. Spectrograms provided in the competition dataset were not used. Instead, new spectrograms were created from the raw EEG data through different techniques, including CWT (Continuous Wavelet Transform) and STFT (Short-time Fourier Transform).
2	Coolz	**Model:** Ensemble of six different models, all using image-based classifiers. **Approach:** Two models using spectrograms, three using raw EEG recordings, and one using a mix of spectrogram and raw EEG. High experimentation of pre-processing techniques and hyperparameter optimization.
3	Dieter	**Model:** Ensemble of sixteen models, including both image-based classifiers and time-series-based classifiers. **Approach:** For the image-based classifiers, CNN architectures pretrained on EEG data were used instead of training from scratch. For the time-series-based ones, the strategy was using 1D convolutions to encode the raw EEG data before modeling.

In general, there was not a single approach that dominated all the others in the competition. The results seem to come from intense experimentation rather than unique insights. Nevertheless, a few key ideas were identified and could be explored to improve the models developed in this project:

1. **Prioritizing high-confidence data:** As mentioned before, the EEG recordings were each annotated by three to twenty experts. The winning teams all had one thing in common: they focused on recordings that had been annotated by a higher number of experts. For example, some competitors trained the model considering only the EEG recordings that had received votes by ten or more experts. There are significant inconsistencies in the opinions of experts to classify a large part of the dataset. As a result, training the model using the data with a lower number of votes actually decreased model performance.

2. **Training deeper models locally:** Another limitation encountered during this project was the Kaggle workspace's memory constraints, which restricted our ability to explore models with extensive computational requirements. Notably, many of the winning entries circumvented this issue by training more complex models, such as EfficientNetB5, on local systems and then executing only the inference processes on Kaggle. This strategy effectively bypassed the memory limitations and facilitated the use of advanced model architectures that could potentially yield better

performance. Moving forward, we recommend adopting this local training approach to enable the incorporation of more sophisticated models into our project.

3. **Developing even more models:** Initially, our project proposed a simple ensemble comprising two distinct models. However, analysis of competition strategies revealed a trend towards employing larger ensembles that integrate numerous models, each developed through extensive experimental trials. This approach not only diversifies the predictive capabilities of the overall system but also enhances its robustness and accuracy. To enhance our project's effectiveness, we should consider expanding our ensemble to include a broader array of models, incorporating varied pre-processing techniques and potentially experimenting with different architectural frameworks.

In summary, we conclude that the project has been successful and that an ensemble approach to EEG analysis has the potential to achieve high performance in identifying harmful brain activity and improve patient care. Multiple areas of improvement have been identified based on methods developed by other competitors and these ideas are recommended for exploration as next steps of this project.

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